

GSC-F: Nonlinear Shock Control

Derived from the Dynamic Field

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Abstract

This paper presents the GSC-F control architecture as a nonlinear algorithm for suppressing extreme transient shocks. The core idea of the algorithm is derived from our dynamic field: the linear stabilizing component is extended by a nonlinear barrier and damping mechanism that is activated only at larger deviations, thereby selectively strengthening the defensive response of the system. The benchmark shows that GSC-F initially significantly outperformed a standard PD baseline, and this result was subsequently confirmed on an independent set of noise seeds. After fairly strengthening the baseline to the strongest identified linear PD-like controller, GSC-F still retained a majority advantage: 80.0% in maximum deviation, 80.0% in settling time, 73.56% in integrated absolute deviation, and 83.56% in control effort. These results suggest that GSC-F is not merely a cosmetic extension of linear control, but a robust nonlinear alternative for shock-dominated tasks.

1 Introduction

Control of systems exposed to sudden transient disturbances remains a persistent challenge in industry as well as in experimental physical and energy-related applications. Classical linear controllers such as PD or PID are often highly effective near the operating point, but under extreme shock their limitations may become apparent: either they require aggressive tuning that increases sensitivity and control demand, or they leave excessively large peak deviations.

In this work, we present the GSC-F algorithm, whose construction is derived from our dynamic field. The basic idea is as follows: the system does not behave merely as a linear deviation corrector, but as a field with an adaptive defensive response. At small deviations, regulation operates in a linear regime. At larger deviations, however, a nonlinear barrier term and nonlinear damping are activated, together producing an enhanced stabilizing response. This models a situation in which the dynamic field reacts weakly in the calm regime but sharply amplifies its intervention when destabilization threatens.

The goal of this work is not to claim universal dominance across all scenarios. The goal is to rigorously verify whether an architecture derived from the dynamic field provides a real performance benefit against fairly chosen linear baseline controllers.

2 Construction of the GSC-F Algorithm

The proposed controller can be written in the form

$$u_{\text{total}} = u_{\text{lin}} - u_{\text{gsc}}, \quad (1)$$

where the linear part is

$$u_{\text{lin}} = K_p(-x) - K_d\dot{x}, \quad (2)$$

and the nonlinear GSC-F term is activated only after a certain amplitude barrier is exceeded:

$$u_{\text{gsc}} = w(|x|) \left(B_{\text{max}} x^3 + K_v |x| \dot{x} \right). \quad (3)$$

The function $w(|x|)$ represents the smooth activation of the nonlinear regime. At small deviations it is practically zero, while at larger deviations it rises toward one. In our interpretation, this corresponds to the activation of a defensive layer of the dynamic field: the linear part provides standard stabilization, while the nonlinear spring-like and damping contribution intervenes only when the system leaves the safe regime.

3 Benchmark and Methodology

The tests were carried out on a simple dynamical model with external shock forcing and stochastic noise. The benchmark included multiple shock amplitudes, multiple shock durations, and multiple noise seeds. Four metrics were evaluated:

- maximum absolute deviation,
- settling time after shock,
- integrated absolute deviation,
- total control effort.

The comparison procedure was performed in three stages:

1. GSC-F vs. a basic PD baseline,
2. independent confirmation on a different set of noise seeds,
3. GSC-F vs. the strongest identified linear PD-like baseline after fair retuning.

4 Results

The first benchmark against the basic PD controller showed a very strong advantage for GSC-F. This result was also confirmed in an independent run with different noise seeds, ruling out the possibility that it was merely a one-off artifact. Subsequent retuning of the linear baseline showed that the strongest identified linear variant had the character of a more aggressive PD-like controller. Against this strengthened baseline, GSC-F no longer exhibited absolute dominance, but it retained a majority advantage across the benchmark space.

Table 1: Performance of GSC-F against the strongest tested PD-like baseline.

Metric	Win (%)	Mean (%)	Median (%)	Min (%)	Max (%)
Peak deviation	80.00	13.99	25.83	-45.45	44.57
Settling time	80.00	10.60	16.19	-20.24	28.01
Int. abs. deviation	73.56	12.63	24.63	-46.97	45.08
Control effort	83.56	2.75	2.53	-1.83	6.63

The table shows that GSC-F maintains a majority advantage in all four metrics. The greatest practical benefit appears in maximum deviation and settling time, which are typically the most important quantities under extreme shock conditions. The improvement in control effort is smaller, but still positive in both the mean and median.

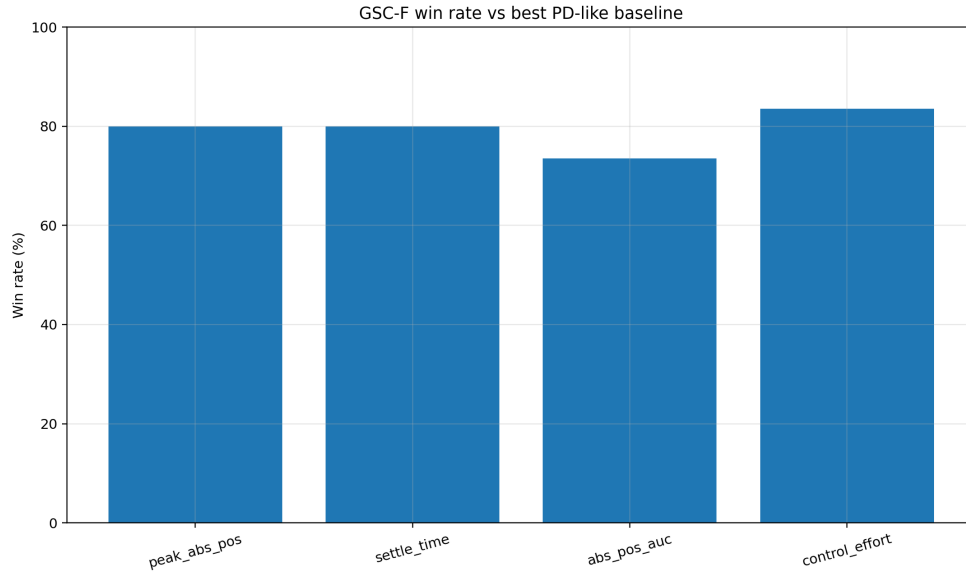


Figure 1: Win rate of GSC-F relative to the strongest tested linear PD-like baseline.

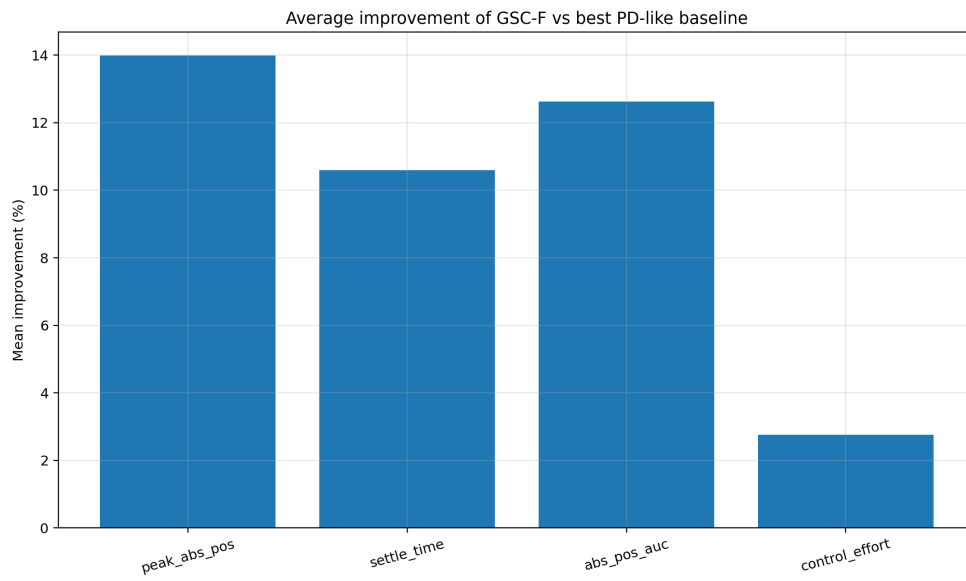


Figure 2: Average percentage improvement of GSC-F across the four main metrics.

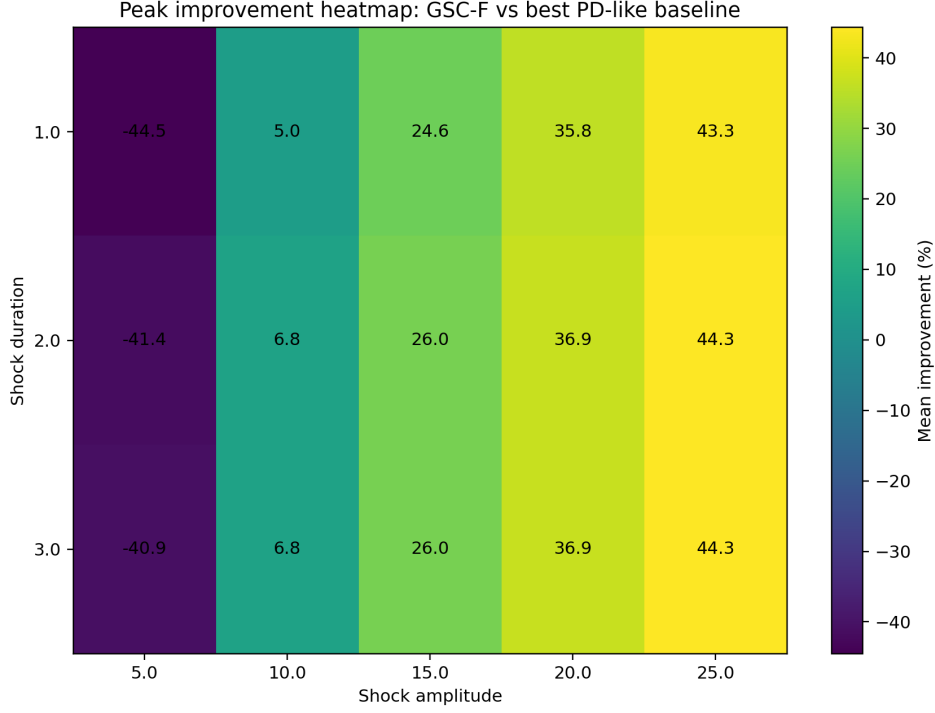


Figure 3: Heatmap of the mean improvement in maximum deviation as a function of shock amplitude and shock duration.

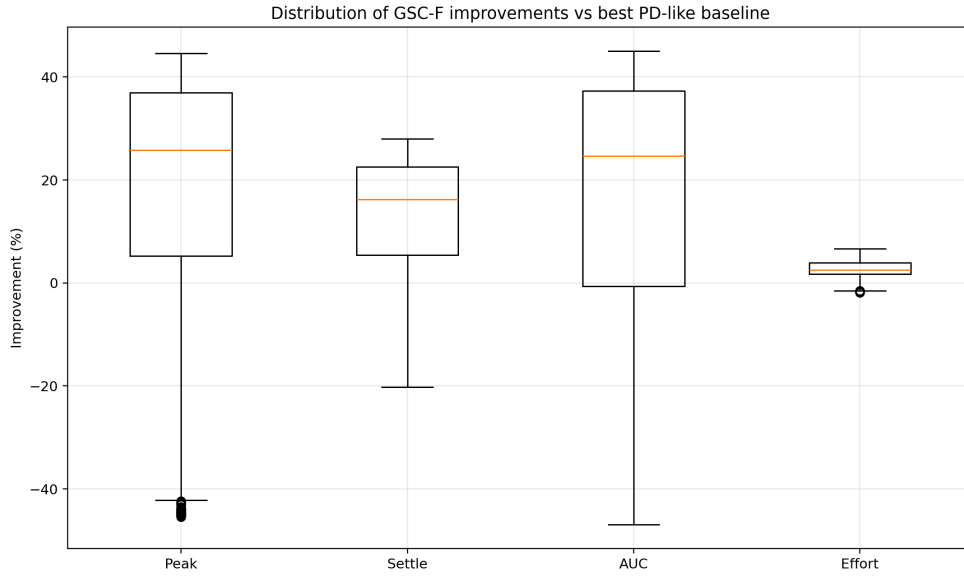


Figure 4: Distribution of scenario-wise improvements. The advantage of GSC-F is majority-based rather than universal.

5 Contribution and Interpretation

The main contribution of this work consists of three points. First, the GSC-F algorithm is not designed as a purely empirical trick, but is derived from our dynamic field concept, in which the defensive response of the system is amplified nonlinearly when leaving the calm regime. Second, the results show that this nonlinear architecture retains a real benefit even after the benchmark is

made stricter. Third, the methodology demonstrates a proper way of evaluating new algorithms: first identify potential, then confirm on a new set of seeds, and finally strengthen the baseline fairly.

6 Limitations of the Study

The present study is based on a simplified dynamical model with shock forcing and stochastic noise. The results therefore cannot be directly transferred to all physical, industrial, or multiphysical systems without further validation.

7 Discussion

The results do not show absolute dominance of GSC-F. This is important to acknowledge openly. In some scenarios, an aggressively retuned linear baseline achieves better results. Nevertheless, GSC-F maintains a majority advantage and remains competitive even after the comparison is made more rigorous.

8 Conclusion

We have presented a publication-oriented benchmark of the GSC-F algorithm, whose construction is derived from our dynamic field. GSC-F first significantly outperformed the basic PD baseline, this result was reproduced on independent seeds, and it was finally tested against the strongest identified linear PD-like baseline. In this final and fair form, GSC-F preserved a majority advantage in all main metrics. This supports the conclusion that it is a robust and meaningful nonlinear architecture for the control of extreme transient shocks.

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